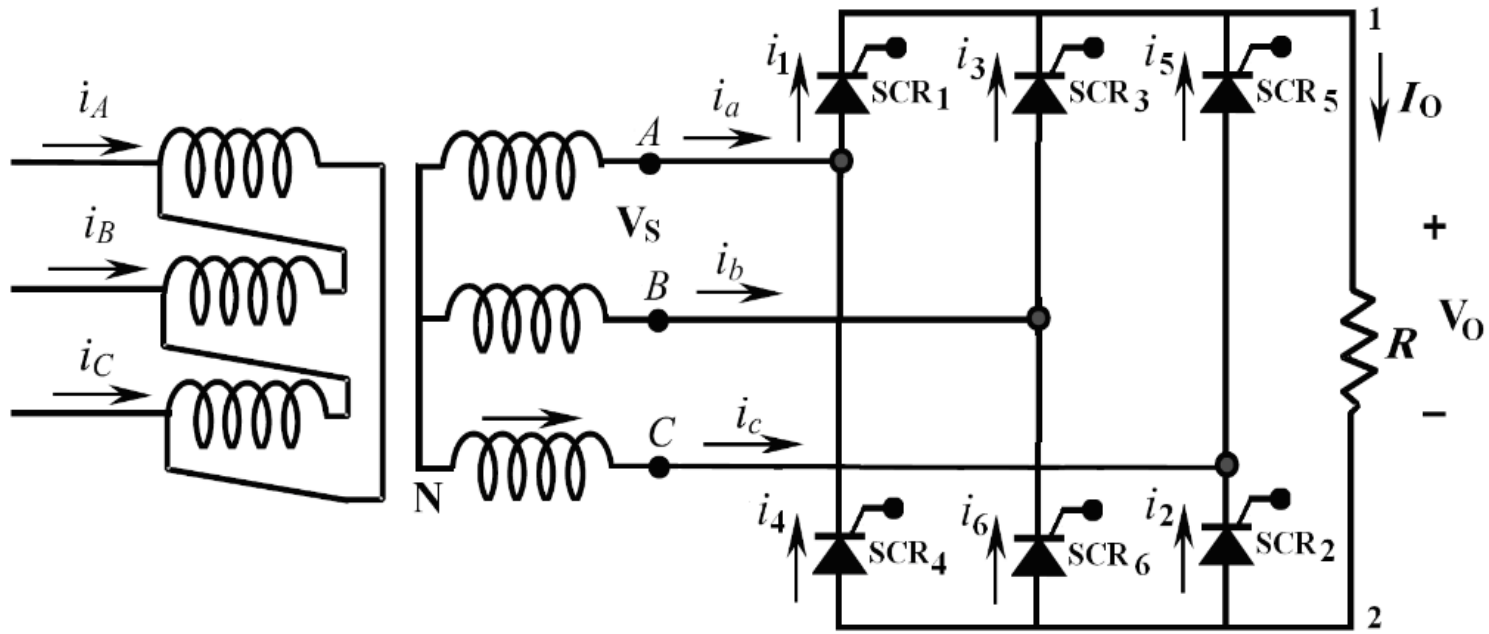
Three Phase Full Wave Controlled Rectifiers

Three Phase Full Wave Controlled Rectifiers

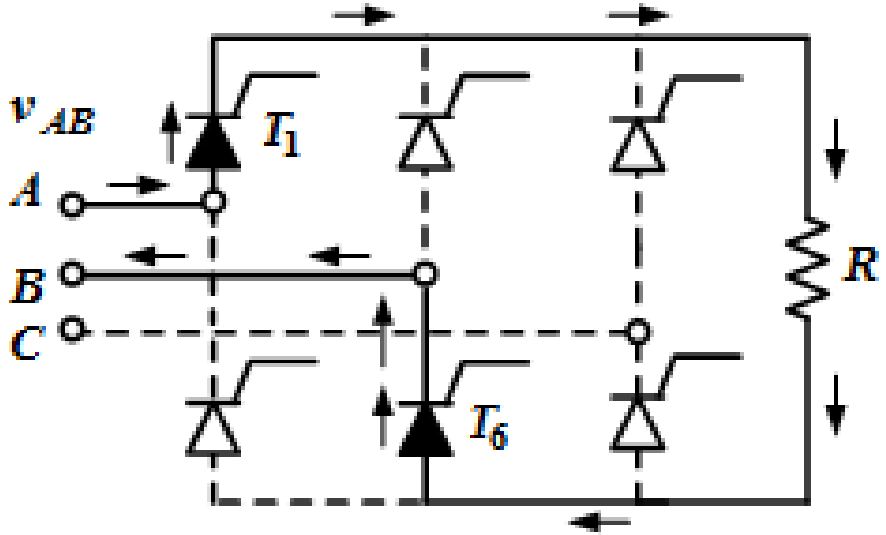
- **Three phase full converter is a fully controlled bridge controlled rectifier using six thyristors connected in the form of a full wave bridge configuration. All the six thyristors are controlled switches which are turned on at a appropriate times by applying suitable gate trigger signals.**


i_A
i_B
i_C
i_a
i_b
i_c
A
B
C
N
V_S
i_1
i_3
i_5
i_4
i_6
i_2
SCR1
SCR3
SCR5
SCR4
SCR6
SCR2
1
2
I_O
+
V_O
R
–

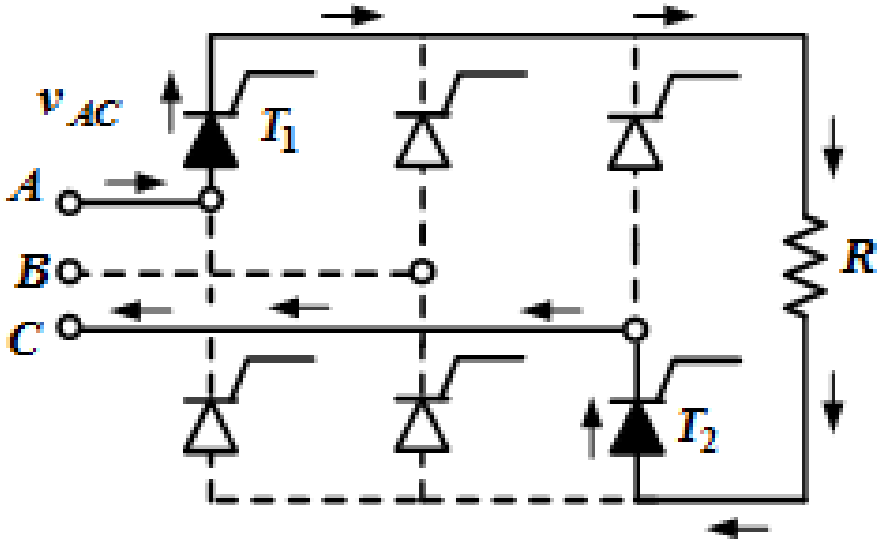
Interval	Voltage at Point 1	Voltage at Point 2	Voltage V_{12}
$0-60°$	A	B	AB
$60-120°$	A	C	AC
$120-180°$	B	C	BC
$180-240°$	B	A	BA
$240-300°$	C	A	CA
$300-360°$	C	B	CB
$360-420°$	A	B	AB

Interval	V_{SCR1}	V_{SCR3}	V_{SCR5}	V_{SCR4}	V_{SCR6}	V_{SCR2}
$0-60^{o}$	0	BA	CA	BA	0	BC
$60-120^{o}$	0	BA	CA	CA	CB	0
$120-180^{o}$	AB	0	CB	CA	CB	0
$180-240^{o}$	AB	0	CB	0	AB	AC
$240-300^{o}$	AC	BC	0	0	AB	AC
$300-360^{o}$	AC	BC	0	BA	0	BC

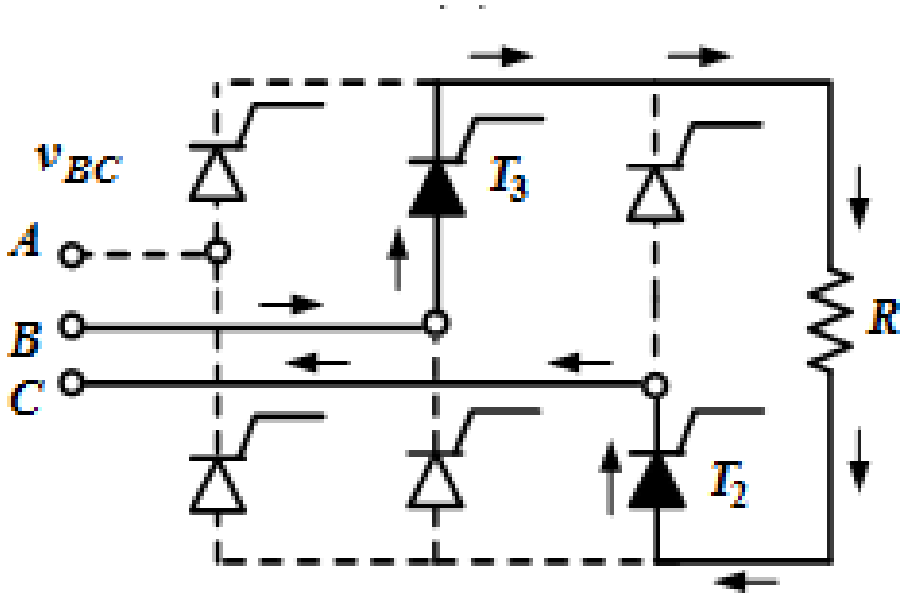
Interval	V_{SCR1}	V_{SCR3}	V_{SCR5}	V_{SCR4}	V_{SCR6}	V_{SCR2}
$0-60^{o}$	0	BA	CA	BA	0	BC
$60-120^{o}$	0	BA	CA	CA	CB	0
$120-180^{o}$	AB	0	CB	CA	CB	0
$180-240^{o}$	AB	0	CB	0	AB	AC
$240-300^{o}$	AC	BC	0	0	AB	AC
$300-360^{o}$	AC	BC	0	BA	0	BC



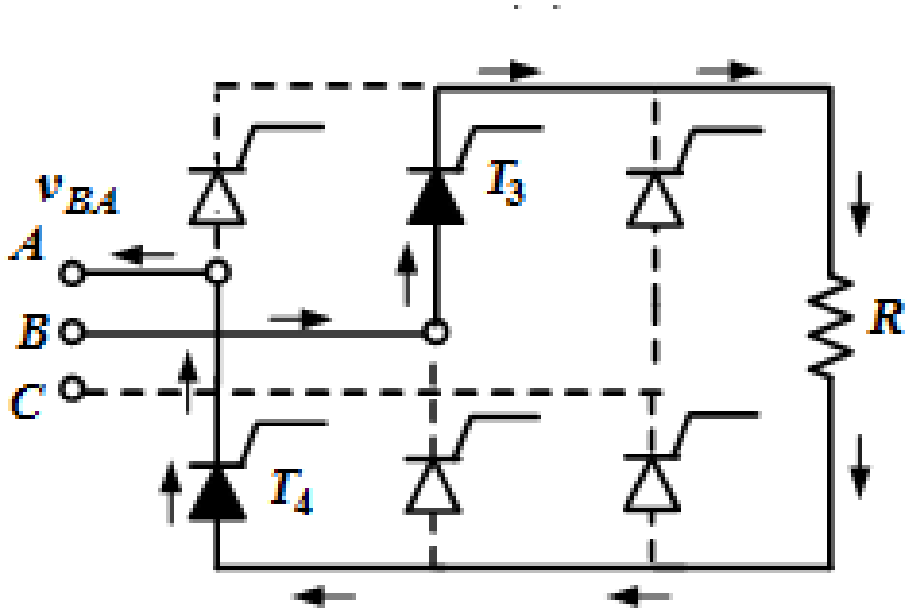
(a)



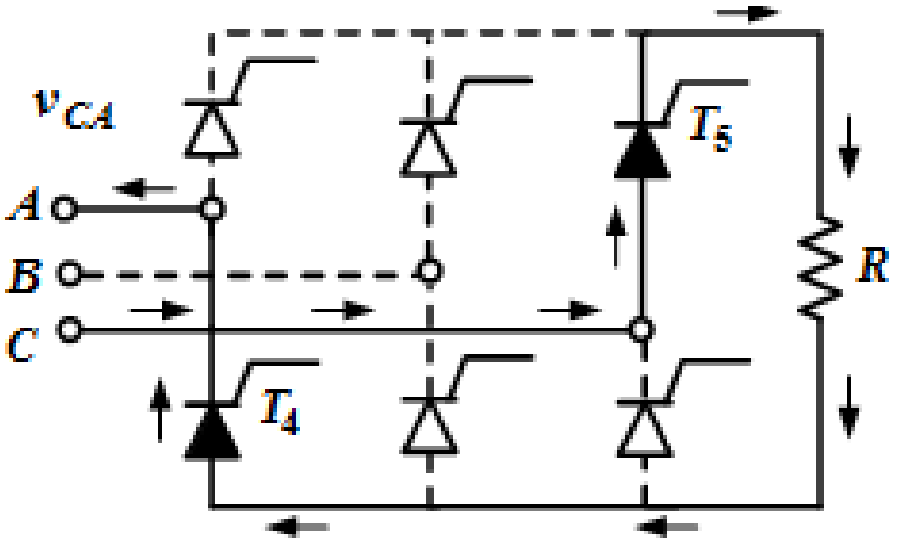
(b)



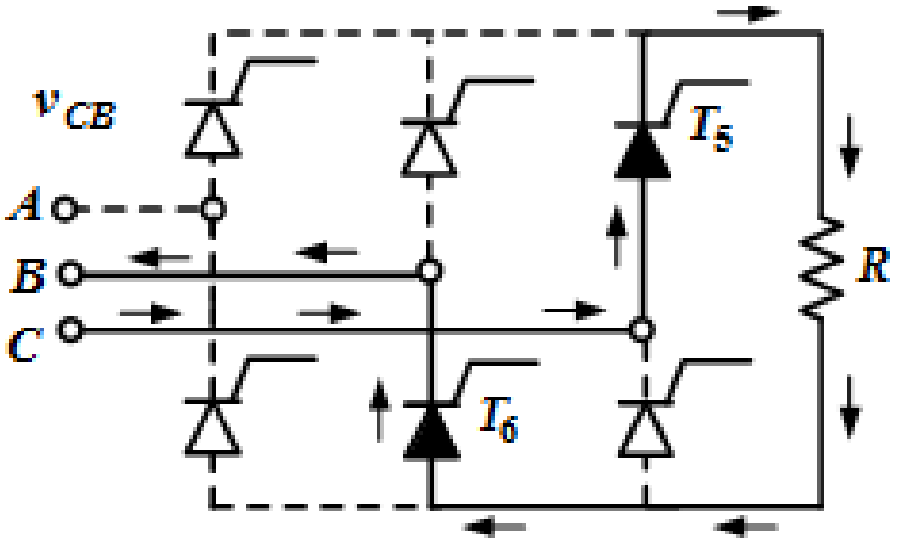
(c)



(d)



(e)



(f)

The average output voltage is given by:

1- Discontinuous mode of operation:

$$V_{dc} = \frac{3\sqrt{3}\,V_m}{2\pi}\left[1 + Cos\,\alpha\right]$$

2-Continuous mode of operation:

$$V_{dc} = \frac{3\sqrt{3}\;V_m}{\pi}\,Cos\,\alpha$$

The rms value of the output voltage is given by:

1- Discontinuous mode of operation:

$$V_{rms} = \sqrt{3}\, V_m \sqrt{\frac{3}{4\pi}\left[\pi - \alpha + \frac{\sin 2\alpha}{2}\right]}$$

2-Continuous mode of operation:

$$V_{rms} = \sqrt{3}\, V_m \sqrt{\frac{1}{2} + \frac{3\sqrt{3}}{4}\cos 2\alpha}$$

The maximum value of the output voltage when $\alpha = 0$ is given by:

$$V_{dm} = \frac{3\sqrt{3}}{\pi} V_m$$

The normalized Average output voltage is given by:

$$V_n = \frac{V_{dc}}{V_{dm}} = \cos \alpha$$

The average SCR current is:

$$I_{Tavg} = \frac{I_{avg}}{3}$$

The rms SCR current is:

$$I_{Trms} = \frac{I_{rms}}{\sqrt{3}}$$